24K-0013

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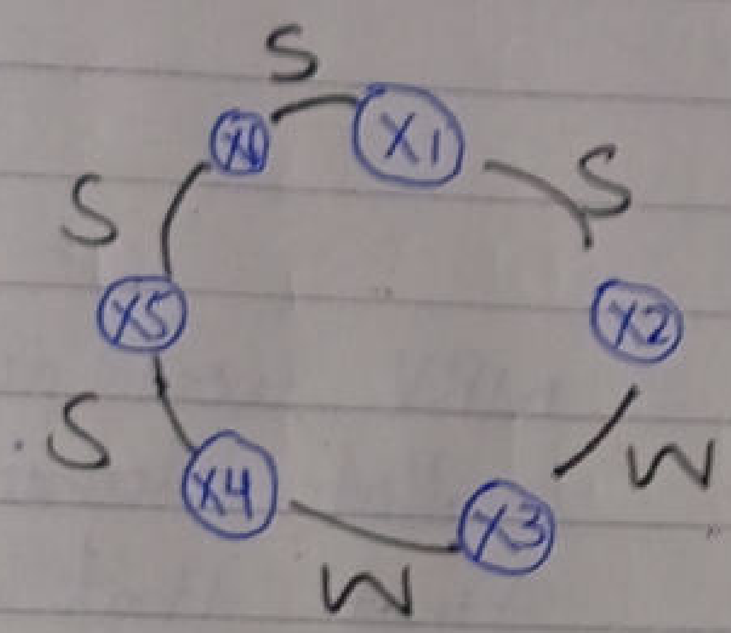
Ai Assignment # 2

Question: 1

a.



Binary:

X1 = P or X2 = P
X2 = E or X3 = E
X3 = E or X4 = E
X4 = P or X5 = P
X5 = P or X6 = P
X6 = P or X1 = P

$\forall i, j$ s.t $Adj(i,j) \rightarrow (X_i = E$ and $X_j = E)$

Unary:

X2 ≠ P
X3 ≠ P
X4 ≠ P

b.

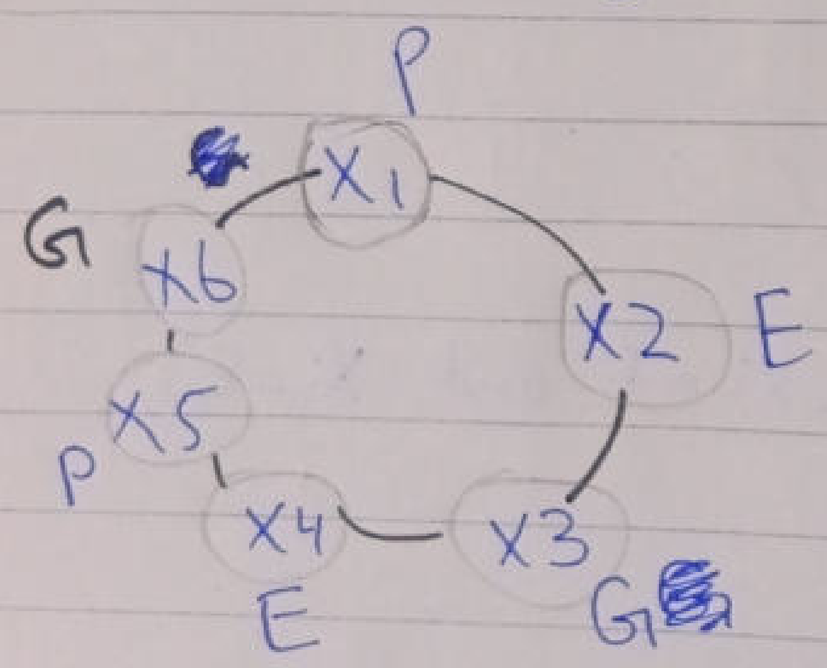
X1			E
X2			
X3		G	E
X4		G	E
X5	P	G	E
X6	P		

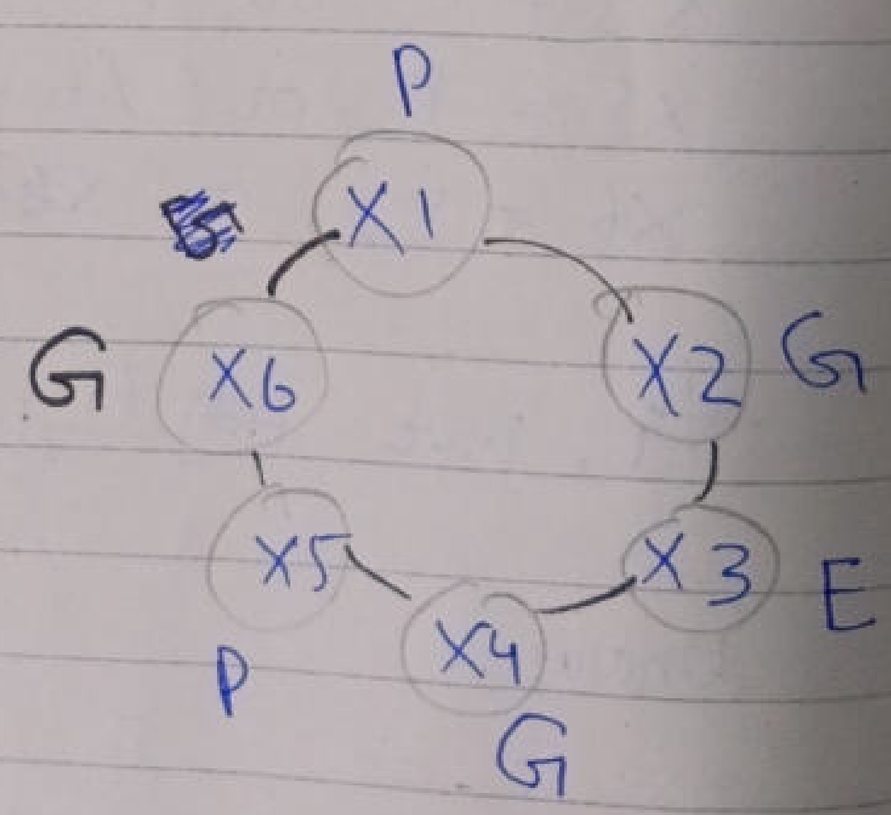
c.

X1			E
X2			
X3		G	E
X4		G	E
X5	P		
X6	P		

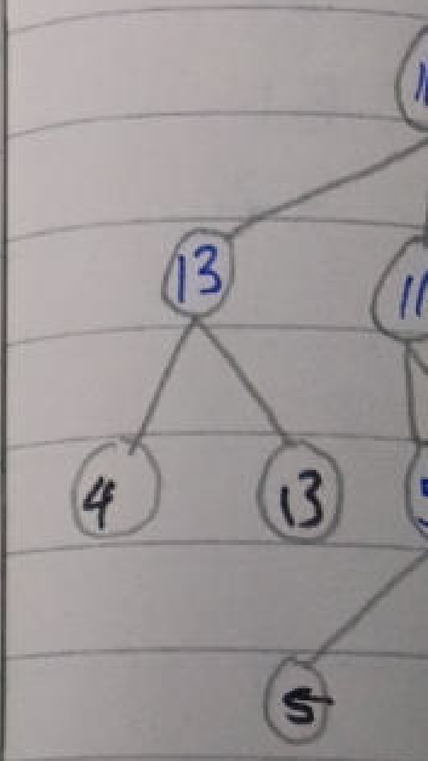
MRV would pick the smallest domain first that is x_1 & x_5

d/e. According MRV : X1 & X5 ties





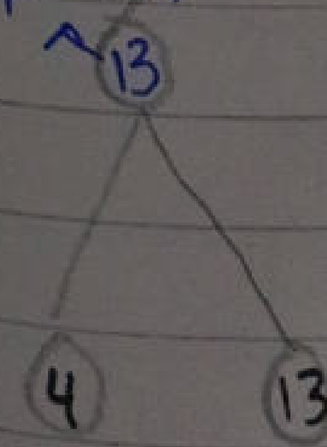
a.



b.

α= -inf, β=

α= 4,13
β= inf



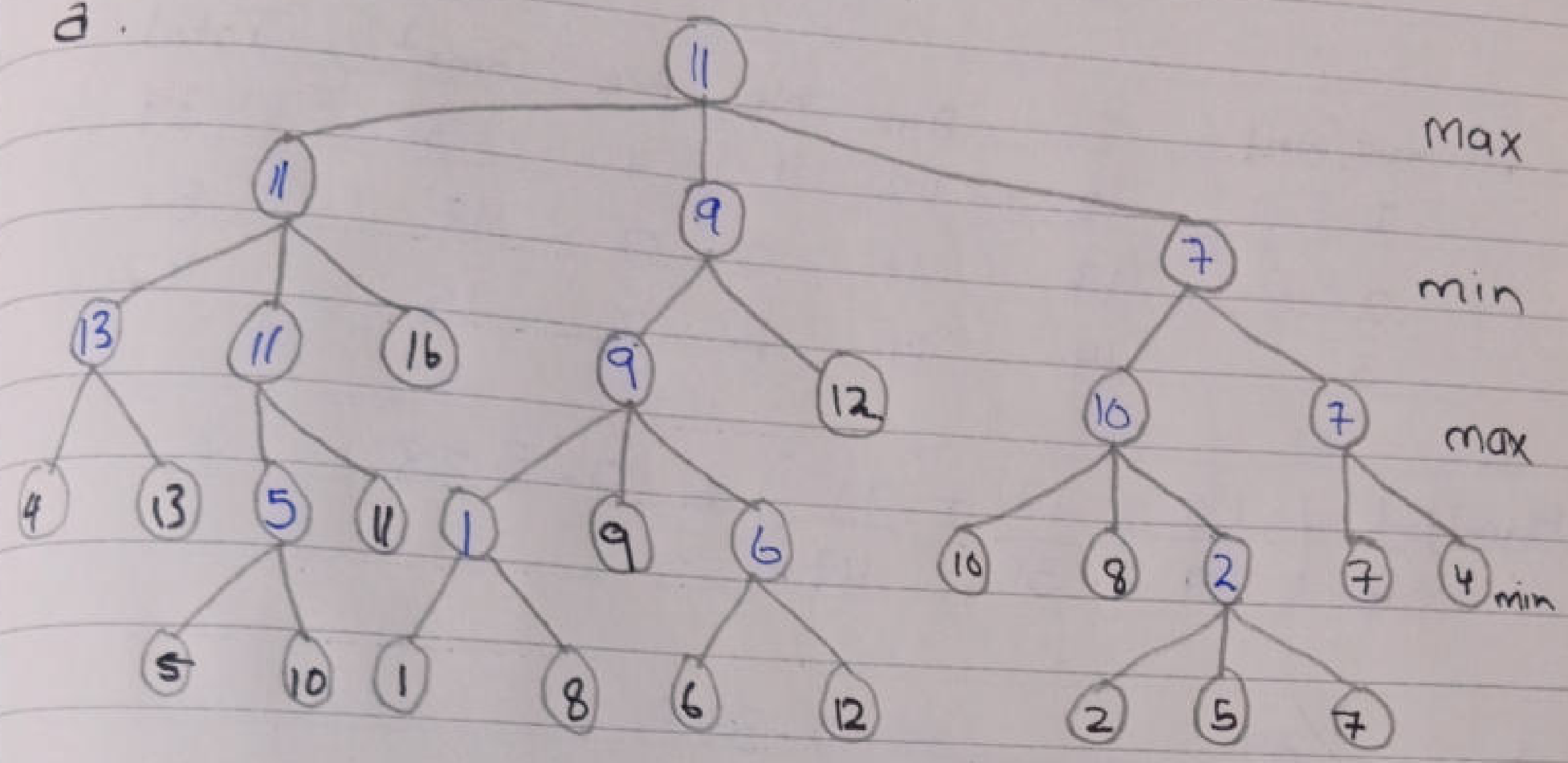
α= -inf
β= 13,5

The

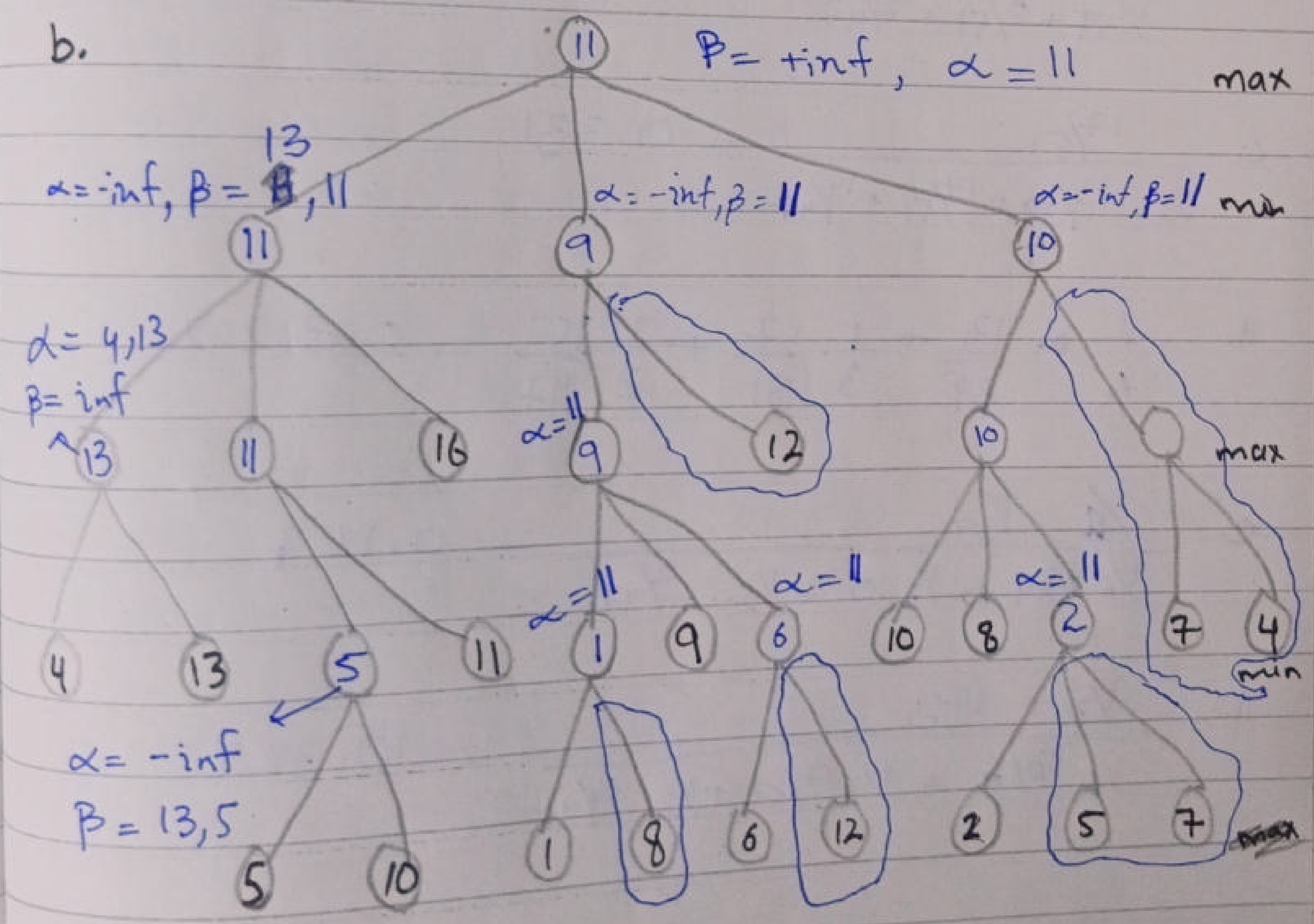
c. No

Question #2

a.



b.



The circle nodes are prunednodes

C. No, The Alpha Beta pruning gives same answer as of minmax

Question # 3

Document/Word	I	Am	We	Are	Groot	Total
1	12	12	1	1	13	39
2	17	17	0	0	17	51
3	14	14	2	2	15	47

a.

$$P(\text{groot}) = \frac{1}{3}\left(\frac{13}{39} + \frac{17}{51} + \frac{15}{47}\right) = 0.328$$

b

$$\frac{1/39}{1/39 + 0/51 + 2/47} = 0.376$$

c

$$\frac{17/51}{13/39 + 17/51 + 14/47} = 0.331$$

d.

$$\frac{1}{6} \cdot \frac{13}{39} + \frac{1}{3}\frac{17}{51} + \frac{1}{2}\frac{15}{47} = 0.3262$$

e.

$$\frac{1/6 \quad 1/39}{1/6 \; 1/39 + 1/3 \cdot 0 + 1/2 \cdot 2/47} = 0.167$$

f.

$$\frac{1/3 \quad 17/51}{1/6 \; 13/39 + 1/3 \; 17/51 + 1/2 \; 16/47} = 0.329$$

Question #4

Spring (S)	¬S
0.25	0.75

S	R	¬R
T	0.7	0.3
F	0.3	0.7

R	Wo	~Wo
T	0.7	0.3
F	0.2	0.8

We	M	~M
T	0.02	0.98
F	0.42	0.68

R	Wo	We	~We
R	Wo	0.12	0.88
R	~Wo	0.25	0.75
~R	Wo	0.1	0.9
~R	~Wo	0.08	0.92

S	Wo	B	~B
S	Wo	0.8	0.2
S	~Wo	0.4	0.6
~S	Wo	0.4	0.6
~S	~Wo	0.4	0.6

a. $P(R) = P(R|S)\,P(S) + P(R|\sim S)\,P(\sim S)$

$= 0.25 \quad 0.7 + 0.3 \cdot 0.75$

$\boxed{= 0.4}$

b. $P(We) = P(We \mid R, Wo)\ P(Wo \mid R)\ P(R)$
$+ P(We \mid \sim R, \sim Wo)\ P(\sim Wo \mid \sim R)$
$R(\sim R) + P(We \mid R, \sim Wo)$
$P(\sim Wo \mid R) P(R) + P(We \mid \sim R, Wo)$
$P(Wo \mid \sim R)\ P(\sim R)$

$\Rightarrow \boxed{0.114}$

c. $R \quad W$

$P(M \mid S) = P(M \mid We) P(We \mid S) + P(M \mid \sim We)$
$P(\sim We \mid S)$

By solving from the table, $P(We \mid S) = 0.1365$

$P(M \mid S) = (0.02 \times 0.1365) + (0.42 \times 0.8635)$
$= \boxed{0.3654}$

d. $P(B \mid S) = P(B \mid S, Wo)\ P(Wo \mid S) +$
$P(B \mid S, \sim Wo)\ P(\sim Wo \mid S)$
$= \boxed{0.6200}$

$P(Wo \mid S)$ is calculated using $= P(Wo \mid R)\ P(R \mid S) +$
$P(Wo \mid \sim R)\ P(\sim R \mid S)$
$= 0.55$

e. $P(S \mid B, \sim W_0) = \dfrac{P(B, \sim W_0 \mid S)\, P(S)}{P(B, \sim W_0 \mid S)\, P(S) + P(B, \sim W_0 \mid \sim S)\, P(\sim S)}$

$= \boxed{0.1875}$

Question # 5

a. It is bad to classify regular mail as spam than it is for spam as regular.

$$P(w_i = 1 \mid C) = \frac{N_{ic} + 1}{N_c + 2}$$

~~P(Study | Spam~~

Word	Regula	Spam
Study	$\frac{3+1}{4+2} = 4/6$	$1/10$
free	$2/6$	$9/10$
money	$2/6$	$5/10$

b. P (Spam | "money for phyc. study")

money = 1 , free = 0 , study = 1

using Naive bayes

$\Rightarrow P(\text{Spam})\, P(\text{study} \mid \text{spam})\, P(\text{money} \mid \text{Spam})\, P(\text{free})$

$\Rightarrow 0.1 \cdot 0.1 \cdot 0.1 \cdot (1 - 9/10) = 1/2000$

$P(\text{reg}) \cdot P(\text{study} \mid \text{reg})\, P(\text{money} \mid \text{reg})\, P(\text{fee} \mid \text{reg})$

$(1 - 0.1) \cdot (2/3)\, (1/3)\, (1/3) = 2/15$

$P(\text{spam} \mid S) = \dfrac{1/2000}{1/2000 + 2/15} \Rightarrow 0.003376$

$\Rightarrow 0.37\%$

c.

$P_x(1/200) = (1-P) \times P(S \mid \text{Regular})$

$\frac{P}{200} = 1-P \; \frac{4}{27}$

$27P = 800 - 800P$

$827P = 800$

$\boxed{P = 0.967}$

Question # 6

Step 1:

(x) independent variable = no. of Crabs

(y) dependent variable = no. of Clams

Step #2:

$\Sigma(x) = 29$ $\quad \{ (\Sigma(x))^2 = 841$

$\Sigma(y) = 1249$

$\Sigma(xy) = 274 + 280 + 360 + 0 + 140 + 0 + 366 + 0 + 253 + 45$

$= 2136$

$\Sigma(x^2) = 4 + 16 + 4 + 25 + 16 + 0 + 9 + 25 + 1 + 9 = 109$

n = No. of cages = 10

Step # 3:

$$m \text{ (slope)} = \frac{n \sum xy - \sum x \sum y}{n \sum (x)^2 - (\sum x)^2}$$

$$= \frac{10 \times 2136 - 29 \times 1249}{10 \times 109 - 841}$$

$$m \text{ (slope)} = -59.68$$

Step # 4:

$$b \text{ (intercept)} = \frac{\sum y - m \sum x}{n}$$

$$= \frac{1249 - 59.68 \times 29}{10}$$

$$b = 297.98$$

Step # 5: eq $\Rightarrow$ $y = mx + b$

Step # 6:

answer = $\boxed{y = (-59.683)\,x + (297.98)}$

Question #7

logistic Regresion model:

$$P(Y=1) = \frac{1}{1+e^{-z}}, \text{ where } z = b0 + b1x1 + b2x2 + b3x3 + b4x4 + b5x5$$

		Co-efficients (b)	New Patient (x) values
2	diabetes	1.5	1
3	Hypertension	1.2	1
4	Smoker	1.0	1
5	Surgery Type	2.0	2
0	Intercept	-8	—
1	Age	0.05	60

$$z = (-8) + (0.05 \times 60) + (1.5 \times 1) + (1.2 \times 1) + (1 \times 1) + (2 \times 2)$$
$$= 2.7$$

$$P(Y=1) = 1 / 1 + e^{-2.7}$$
$$= \frac{1}{1 + 0.067}$$
$$\boxed{= 0.937}$$